Assignment 3 - Nuclear Fund Momentum Based Algorithm

451 – Financial Engineering – Northwestern University

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Project Overview

This project aims to design, implement, and back test an automated algorithmic trading strategy for a thematic Nuclear Energy fund. The fund invests in 11 publicly traded companies across the nuclear energy landscape. The investment thesis of this fund is forecasting rapid electric demand from the influx of AI data centers that can create a multi-year tailwind that will generate large returns for the fund.

The research question is whether a systematic, rules-based trading strategy applied to this universe can deliver competitive risk-adjusted returns relative to passive market benchmarks. The fund is not actively discretionary; every position is determined by an explicit, repeatable algorithm, which makes the strategy fully testable and removes behavioral bias from execution.

Fund Layout

The fund consists of eleven nuclear-sector equities, organized into three sub-sector buckets that diversify the thesis across different economic drivers:

- **Uranium Miners & Fuel —** uranium miners and fuel-cycle companies (CCJ, UEC, NXE, DNN), which provide commodity exposure to the uranium price.
- **Reactor Developers —** next-generation and small modular reactor developers (OKLO, SMR, NNE, LEU), which provide growth exposure to reactor technology commercialization.
- **Nuclear Utilities —** utilities operating nuclear generation capacity (VST, CEG, TLN), which provide regulated, cash-flow-oriented exposure.

The strategy is benchmarked against four buy-and-hold baselines: the S&P 500 (SPY) for broad-market exposure, the NASDAQ-100 (QQQ) for growth-equity exposure, long-term US Treasuries (TLT) for a fixed-income reference, and a uranium/nuclear sector ETF (URA), which serves as the most direct test of whether the strategy adds value over simply owning the sector passively. Lastly the momentum based strategy will be compared against the nuclear equity fund if it maintained only a buy-hold strategy to provide a final verdict if the strategy is effective in optimizing return and risk.

Data Treatment

All market data consists of daily adjusted closing prices retrieved programmatically from Yahoo Finance using the yfinance Python library. Adjusted prices are used so that dividends and stock splits are already incorporated into the return series, which prevents artificial jumps on corporate-action dates. The backtest window runs from January 2018 through December 2025.

Several holdings (OKLO, NNE, TLN) listed partway through the backtest window. Since the trend filter requires 200 days of price history before it can generate a signal, a security that has not yet listed simply cannot be selected, and its weight is zero until sufficient history exists. Within the return calculation, a missing daily return for a held position is treated as a flat (0%) return rather than being silently dropped from the portfolio sum.

Algorithm Design

The strategy follows the systematic trend-following philosophy described in Andreas Clenow's Trading Evolved. It combines a trend filter, a fixed bucket allocation for position sizing, and a buy-and-hold benchmark comparison for evaluation.

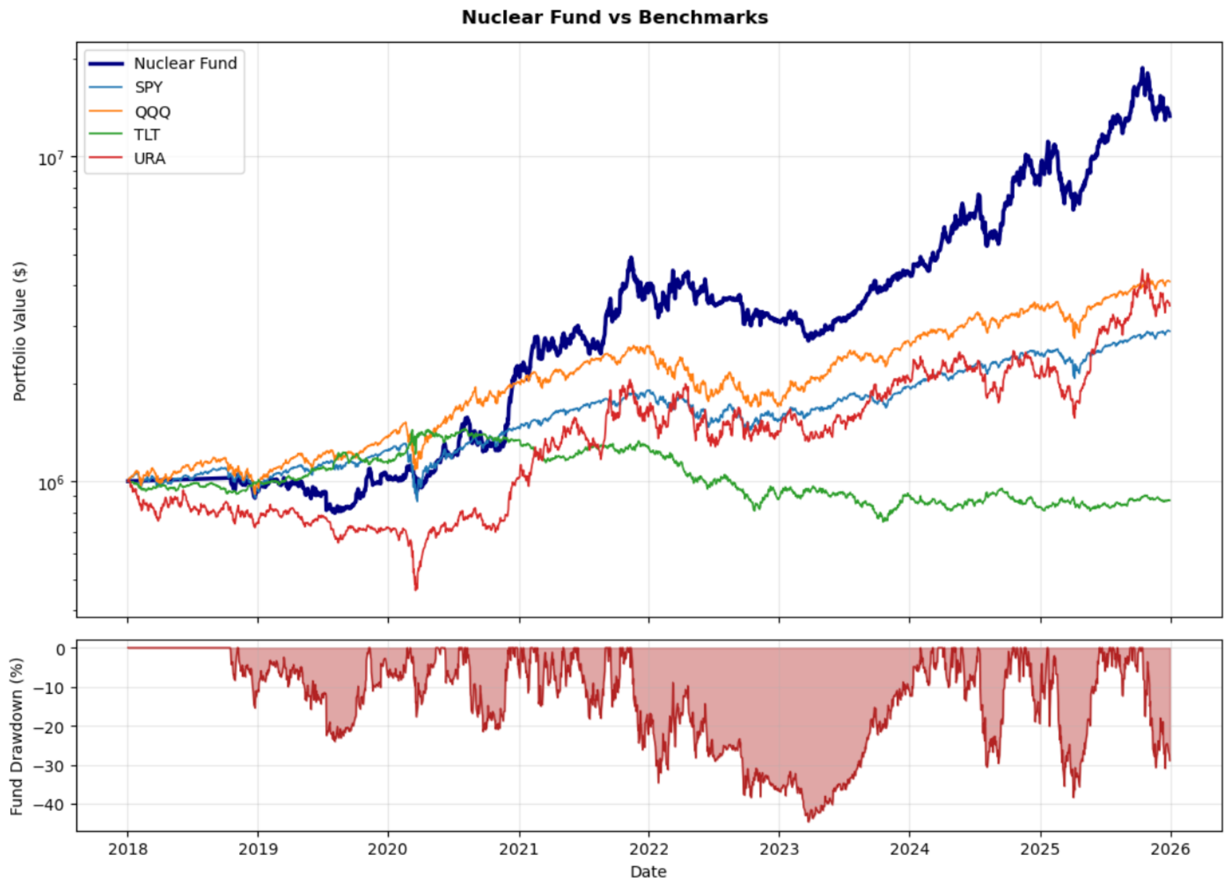
The trading rules for the trend filter are that the stock will be held in the portfolio when its 50 day average exceeds its 200 day moving average - a signal of upward momentum. The allocation is split across 3 classes within the nuclear space – 40% uranium miners equities, 30% reactor development and 30% utilities. Within each bucket, the securities that pass the trend filter share that bucket's target equally. If no security in a bucket passes the filter, that bucket's capital is held in cash. All cash that is not allocated is assumed to be earning the risk free rate of 3%.

Metrics for Evaluation

First and foremost, total return is calculated and compared between the algorithm, benchmarks, and if the securities in the fund were simply held for the duration of the testing. The other significant metric we will evaluate each strategy on is the Sharpe ratio. This will allow us to quantify the return vs risk of the algorithm and benchmarks. Additional metrics for further evaluation and comparison are the Sortino ratio (penalizes only downside volatility, maximum drawdown, and volatility.

Results

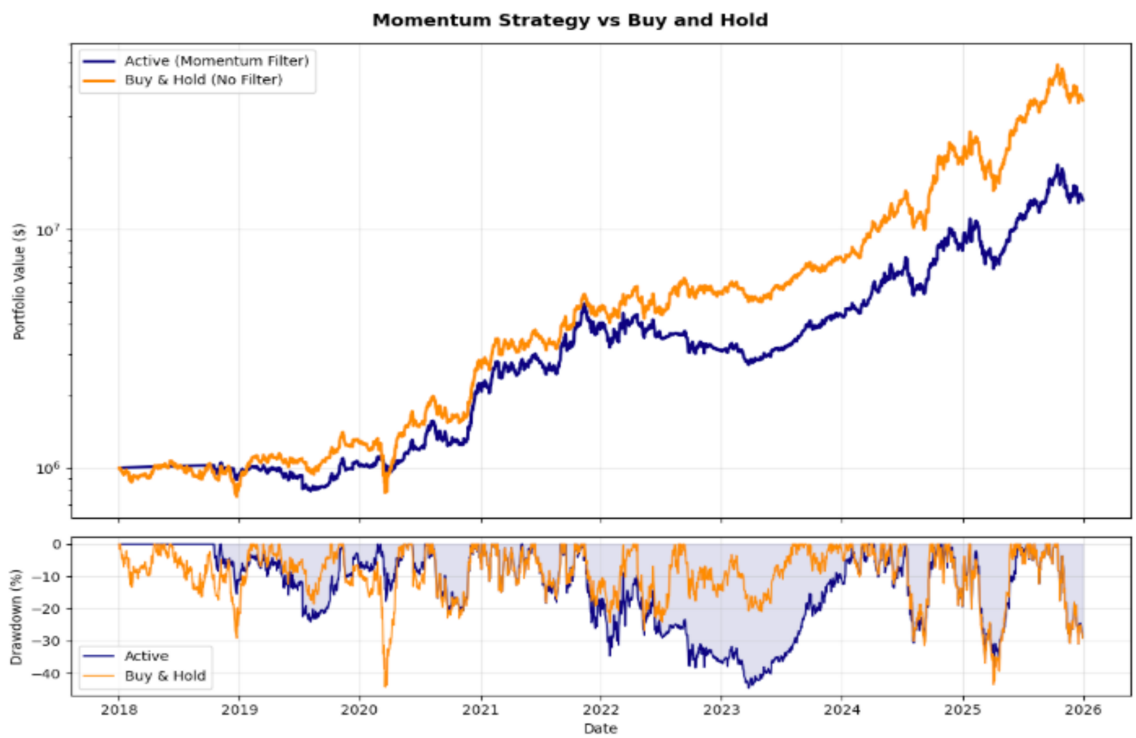
The figure below compares the cumulative growth of the Nuclear Fund using the momentum strategy against the four buy-and-hold benchmarks over the backtest period, on a logarithmic scale. The lower panel shows the fund's drawdown over time.



As shown in the graph above, the Nuclear Fund outperformed all four benchmarks over the period from January 2018 to December 2025. Outperforming Treasuries is expected, but the fund's outperformance relative to QQQ, the S&P 500, and URA is a strong result. Not only is the total return substantially higher, the Nuclear Fund momentum strategy also posts the highest Sharpe ratio of the group. Its volatility is higher than that of QQQ or the S&P 500, but the magnitude of its returns more than compensates for the additional risk. volatility of URA confirms that this is an inherently volatile sector, yet the specific equities selected for the Nuclear Fund significantly outperformed the broad URA index. The Sortino ratio is higher than the Sharpe ratio which indicates that a large amount of the Nuclear fund’s volatility was on the upside, even compared with the URA index.

To further validate the momentum strategy, we compared it against a buy-and-hold version of the Nuclear Fund. The results of the benchmark comparison raised a plausible alternative explanation: that the large returns came simply from exposure to a high-growth sector, rather than from the trading rules imposed on the fund.

The buy-and-hold strategy was evaluated over the same time frame and using the same metrics, and the results are shown below.



This comparison confirmed our suspicion. Simply buying and holding the same stocks from the start of 2018 would have produced larger returns than the automated momentum strategy, indicating that the higher return was driven by stock selection rather than by the strategy itself. This does not devalue the work; rather, it illustrates that a momentum strategy is less applicable to a high-growth sector. These stocks were in the early stages of the recent nuclear renaissance and would generally be regarded as growth stocks which we learned is a profile not well suited to trend-following rules, which are designed to avoid sustained downtrends that simply did not occur during this period.

Further study would focus on an algorithm that capitalizes on the volatility the nuclear sector does exhibit, rather than filtering it out. Tighter momentum rules and additional tools such as the Relative Strength Index (RSI) could enhance the algorithm and produce a more aggressive model aimed at optimizing returns within the nuclear sector.

Conclusion

This project produced a complete, automated trading and backtesting system for a thematic nuclear-sector fund, spanning stock selection, data acquisition, signal generation, position sizing, and risk-adjusted evaluation. Though the fund greatly outperformed outside benchmarks, it was determined that a simple buy and hold strategy would be optimal to the automated trading strategy deployed by the Nuclear Fund. The result showcased that the strong returns came from picking favorable stocks and a favorable sector over the tested period. More broadly, the work demonstrates that the merit of a trend-following overlay is conditional on market regime, and that an honest backtest is one that identifies when a strategy is expected to work, not only whether it worked once. These findings provide a clear foundation for the next iteration of the algorithm, one designed to engage with the nuclear sector's volatility rather than filter it out.